Common-Law Energy and Equivariant Matching for Weak Inhomogeneous Duffin–Schaeffer Approximation

A Scoped Ram Calculus Study of M048

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Abstract

The September 25, 2026 source claims the weak inhomogeneous Duffin–Schaeffer conjecture for every fixed real shift, with unrestricted numerators. This report studies its finite projected-table mechanism, not the validity of the entire arithmetic proof. We preserve the separate mask and numerator laws, the common translation orbit, and the realized adaptive fields in the virtual comparison. An exact residue-flux representation turns the next accounting-energy increment into a convex quadratic matching objective. An equivariant minimizer preserves the source's endpoint capacities, cap and first mass, and a two-class example strictly improves the permitted product coupling. We give exact optimality certificates, loss-sensitive energy bookkeeping, interval Gram checks, and a conditional transfer theorem requiring only uniform integrability of the expected retained arrays. Neither covariance positivity nor finite verification supplies the arithmetic extraction, sparse-alignment or chronological-partition estimates. Private Ram proofs are used only locally, and the current public comparison distinguishes the weak statement from the false coprime analogue. Contribution assessment: C3/5, an equivariant variance-minimizing prime-step construction with a strict finite improvement and a proved weaker analytic transfer interface, without a new global arithmetic theorem.

1 Problem, status, and evidence

Work on $\mathbb{T} = \mathbb{R}/\mathbb{Z}$ with normalized Lebesgue measure m. For a fixed $\gamma \in \mathbb{R}$ and finite-valued $\psi : \mathbb{N} \to [0, \infty)$, set

$$r(q) = \psi(q)/q, \qquad E_q = \{x \in \mathbb{T} : \|qx - \gamma\|_{\mathbb{R}/\mathbb{Z}} < \psi(q)\}. \tag{1}$$

The source [1] claims that $\sum_q \phi(q) r(q) = \infty$ implies $m(\limsup_q E_q) = 1$. The exceptional null set may depend on both γ and ψ. There is no numerator coprimality restriction, monotonicity assumption, or simultaneous assertion for all shifts outside one common null set. The source's divergence statement is not a convergence criterion for the unrestricted sets E_q.

What is checked here. The complete projected-tables section and elementary conventions were literally read, together with the introduction, selected scientific passages in every other main proof branch, and the complete final Hausdorff argument. Sections below give self-contained proofs of the local constructions. The arithmetic inputs remain imported source assertions: this is not an independent proof of the conjecture. Exact line intervals, unread complements and SHA256 hashes are in `M048-read-scope.json`; heading searches receive no literal-read credit. The source snapshot's separate parent authentication is provenance, not author proof evidence.

Current primary comparison. Koukoulopoulos and Maynard proved the homogeneous Duffin–Schaeffer theorem [2]. Beresnevich, Hauke and Velani's Theorem 14 [3] treats rational $\gamma = A/B$ with the adapted condition $(A + aB, q) = 1$, and hence the weak rational-shift statement. Their Theorems 11–12 connect uniformly distributed selected numerators and second-moment bounds to full measure. These are antecedents, not new Ram results. The September 2026 preprints [4, 5] claim counterexamples with $(a, q) = 1$ for every nonzero rational shift; [5] also claims a residual set of bad shifts. These do not refute the unrestricted-numerator weak claim. The refresh inspected primary statements, not these papers' complete proofs, and does not establish exhaustive novelty or community acceptance of the source.

1.1 Source architecture and retained details

Source module	Role and boundary of this report
Introduction; preliminaries	Fixed shift, small dyadic widths, late finite blocks, deficit probability law and determinant-cycle means. Read complete.
Centres	Popular multiples, assignment density, raw-row alternative, prime-height and parameter hierarchy read; host adjustment and exterior decay are imported.
Prescriptions	Good masks, first projection and the complete soft-loss proof read; detailed hard rational-model prescriptions are imported.
Tables	Complete definitions, prime-step proof, energy argument, virtual marginals, discard telescoping, equidistribution and conditional reduction read. Principal local worksite.
Arithmetic; direct	Small-scale extraction statement and initial proof; direct-birth statement and determinant coordinates read. Remaining inverse and box-counting proofs imported, not certified by a Gram identity.
Sparse	Full sparse-step statement and its hypotheses read; prime-window, switching and train-coverage proofs imported.
Partitions	Target-grid count and mask-moment proofs, history-test definition and start of its cost proof read. Remaining cost proof and operational partition construction imported.
Joins	Later-omission averaging, capacity/unique-charge proof and linked main-term restoration read; full final chronological closure read. Other backward-expansion and roundoff estimates imported.
Consequences; figures	Complete Hausdorff deduction and both structural figure sources read. Mass transference is an external theorem, not reproved.

The source first handles $qr(q) \not\to 0$ using uniformly spaced grids. Otherwise widths can be

decreased to dyadic values and late finite blocks chosen with $w \le W = \sum_q \phi(q) r(q) \le 2w$. The construction assigns rows to centres, fixes masks and birth tests, then proceeds through partitions, cap clipping, prime matching and class coarsening. A deleted connector does not split an inherited accounting class. Stored history, not future partitions, supplies later tests.

The three major inputs in `tables.tex` are first-sharing comparisons, sparse-alignment costs and chronological partitions. Their uniformity in prior allowed histories is essential. In particular, the narrow clipping constants must be fixed before the cap N_0. After all loss and energy constants are fixed, the final reduction chooses one finite union approximating the avoided set; only then does the least denominator tend to infinity. We retain this order below.

2 The actual Ram interface

The private original [6], Proposition `prop:psd`, proves that square-integrable features under one specified probability law have $M = \mathbb{E}(\varphi\varphi^T) - \mathbb{E}\varphi\,\mathbb{E}\varphi^T \succeq 0$ and $\mathrm{Cov}(B\varphi) = BMB^T$. Its proof is the integral of a square, not an independence hypothesis. Theorem `thm:remainder` there keeps both cross covariances in a common-law approximation. The private original [7] proves the finite factorization $K = C^T W C$ and keeps the cross term in a quarter-scale energy expansion; its asymptotic RH discussion expressly does not follow from positivity. These passages, not just an index, were read locally. No private title or proof text was sent to public search.

Here the features are interval incidences or residue-success indicators, not smooth polynomial approximations. The same finite Gram construction is exact on a declared law. Merely renaming it "Ram" is not a contribution; the change of the source's matching choice in Section 4 is the substantive local use.

For retained and virtual sums $0 \le F \le Z$ on the *same* space, write $D = Z - F$. Whenever second moments exist,

$$\begin{aligned} \int F^2 &= \int Z^2 - 2\int ZD + \int D^2, \\ \mathrm{Var}(F) &= \mathrm{Var}(Z) - 2\,\mathrm{Cov}(Z, D) + \mathrm{Var}(D). \end{aligned} \tag{2}$$

No independent-noise interpretation of D is valid. The source bounds the retained second moment; it need not bound the virtual second moment. Small L^1 deletion by itself provides neither an L^2 deletion bound nor uniform integrability: $D_n = n\mathbf{1}_{[0,n^{-2}]}$ has integral $1/n$ and second moment one, and larger amplitudes can make the second moment diverge. The transfer theorem below does not impose a virtual L^2 bound.

3 Currencies and a one-prime model

For a centre L, the deficit law is

$$\pi_L(d) = \frac{\phi(L/d)}{L} = \frac{R(L/d)}{d}, \qquad d \mid L, \quad R(n) = \phi(n)/n. \tag{3}$$

It sums to one. At a current remaining part I and mask $a \mid I$, the source's label weight is $\lambda_x = \pi_I(a) w_x$, with $0 \le w_x \le 1$. First mass is $\sum_x r_x \lambda_x$ per unit length. Broad pairs carry $\min(r_x, r_y)\lambda_x\lambda_y$; same-mask clipping pairs carry $r\pi_I(a) w_x w_y$, with only one $\pi_I(a)$. A mask law is not a numerator divisibility probability.

Fix one prime p, one exact width $r > 0$, and one transported cell. Several after-prime masks h may occur. For mask h let $\pi_h = \pi_{I'}(h) > 0$, plus raw loads u_i and minus raw loads v_j. Indices from different masks cannot be matched. The plus load includes the source's residual scalar R_p if the centre's exponent is at least two. Thus before-prime weighted endpoint masses are $\pi_h u_i/p$ and $\pi_h R_p v_j$, where $R_p = 1 - 1/p$. Assume each list load is at most N_0.

A matching $X = (x_{ij})$ for each mask satisfies

$$x_{ij} \geq 0, \qquad \sum_j x_{ij} \leq u_i, \qquad \sum_i x_{ij} \leq v_j, \qquad \sum_{ij} x_{ij} = \min(P_h, M_h), \tag{4}$$

where $P_h = \sum_i u_i$, $M_h = \sum_j v_j$. The zero-load cases have zero matching. Otherwise product coupling $x_{ij} = u_i v_j / \max(P_h, M_h)$ is feasible.

One common orbit. Use a uniform $g \in \mathbb{Z}/p\mathbb{Z}$ and fixed colors $c_i \in \mathbb{Z}/p\mathbb{Z}$, with $S_i(g) = \mathbf{1}_{g=c_i}$. For the source, translation by $1/p^\nu$ changes each plus numerator by a unit modulo p, so its success set has exactly this form after solving the linear congruence. A full p^ν orbit repeats each color equally. All matching data are transported with the cell, not chosen separately after observing success. In particular

$$\mathbb{E}S_i = 1/p, \qquad \mathrm{Cov}(S_i, S_k) = \mathbf{1}_{c_i=c_k}/p - 1/p^2. \tag{5}$$

Equal colors are perfectly correlated; distinct colors cannot both succeed. Replacing these indicators by independent Bernoulli variables changes the law and generally the energy.

Lemma 3.1 (All feasible matchings preserve the local invariants)**.** *For every matching satisfying* (4)*, the post raw loads are*

$$u_i' = u_i S_i, \qquad v_j' = v_j - \sum_i x_{ij} S_i - \frac{v_j - \sum_i x_{ij}}{p}. \tag{6}$$

They are nonnegative and do not exceed their respective incoming raw loads. For each mask their total is at most $\max(P_h, M_h) \leq N_0$*. Their weighted orbit means equal the before-prime weighted endpoint masses.*

Proof. Rewrite v_j' as $\sum_i x_{ij}(1 - S_i) + R_p(v_j - \sum_i x_{ij})$. This proves both endpoint bounds. Each matched portion is present exactly once between its two endpoints. Because the total matching saturates the smaller list, unpaired portions lie on only one side; each can only decrease. The total cap follows. Averaging (6) uses only $\mathbb{E}S_i = 1/p$, giving $\mathbb{E}u_i' = u_i/p$ and $\mathbb{E}v_j' = R_p v_j$. Multiplication by $r\pi_h$ proves first-mass preservation. □

4 Residue-flux optimization of the source matching

Let $\mathcal{A}$ be the old accounting classes within the cell, including classes whose endpoints may vanish. Never split a class when deleting a label. Write A_i^+ and A_j^- for endpoint classes. In $\mathbb{R}^{\mathcal{A}}$, with standard basis e_A, define

$$d_i(X) = \pi_h \left(u_i e_{A_i^+} - \sum_j x_{ij} e_{A_j^-} \right), \quad z_c(X) = \sum_{i:c_i=c} d_i(X), \quad \bar{z}(X) = \frac{1}{p} \sum_c z_c(X). \tag{7}$$

The sum in z_c runs over all masks in the cell. Consequently it retains cross-mask covariance even though matching cannot cross masks.

Theorem 4.1 (Exact flux objective and zero-cost criterion)**.** *Let m_A be the before-prime weighted mass of class A and $Y_A(g)$ its post mass. Then*

$$Y(g) - m = z_g(X) - \bar{z}(X), \tag{8}$$

$$J(X) := r \sum_A \{\mathbb{E}Y_A^2 - m_A^2\} = \frac{r}{p} \sum_{c=0}^{p-1} \|z_c(X) - \bar{z}(X)\|_2^2.$$

In particular J is a nonnegative convex quadratic on the product of the matching polytopes. Its minimum is zero exactly when the linear equations $z_c(X) = z_0(X)$ for all c have a feasible solution.

Proof. Subtract the old masses in (6). Plus endpoint i gains $\pi_h u_i(S_i - 1/p)$ and minus endpoint j loses $\pi_h \sum_i x_{ij}(S_i - 1/p)$. Summing within classes gives $Y - m = \sum_i d_i(S_i - 1/p) = z_g - \bar{z}$. This vector has mean zero, so expansion of each square leaves its variance. This proves (8). It is the square of an affine map in X, hence convex. A finite sum of nonnegative squares vanishes exactly when every square does, proving the feasibility criterion. □

Equivalently, let D have columns d_i. Then

$$\mathrm{Cov}(Y) = D \left[\frac{\mathbf{1}_{c_i = c_k}}{p} - \frac{1}{p^2} \right]_{i,k} D^T. \tag{9}$$

This is a concrete application of the private common-law theorem, but the identity itself is standard finite covariance algebra. The benefit is that its coefficients depend on a genuine choice in the source mechanism.

Theorem 4.2 (Equivariant minimum-variance matching)**.** *On fixed finite incoming data, minimize J over (4) for all masks jointly. Among its minimizers, minimize $\sum_{ij} x_{ij}^2$. This defines one unique matching X_*, preserves all conclusions of Lemma 3.1, and satisfies $J(X_*) \le J(X_{\mathrm{prod}})$. If a symmetry permutes endpoint and class labels, preserves mask weights and loads, and permutes all residue colors by one common permutation, the chosen matching transports equivariantly under that symmetry.*

Proof. The feasible set is nonempty by product coupling, closed and bounded, hence compact. Continuity gives a minimizer. Convexity of J makes its minimizer set convex; it is also compact. The squared Euclidean norm is strictly convex, giving a unique secondary minimizer. The symmetry maps feasible matrices bijectively to feasible matrices, preserving both objectives. Uniqueness therefore forces transport of the chosen matrix. The invariants hold for every feasible matrix, and product coupling is one such matrix, proving the remaining assertions. □

Why the source's core symmetry is compatible. For current core $C = p^\nu H$, with $(H, p) = 1$, a plus projected denominator V_i is divisible by C and has p-valuation ν. Translation by $1/p^\nu$ has numerator increment V_i/p^ν; translation by $1/C$ has increment V_i/C. Solving the success congruence shows that the latter translation shifts every color by $-H^{-1} \pmod p$, independently of i. Other data transport by their label permutations. The optimizer therefore does not privilege the realized success color. It is defined from the complete current orbit and past data, not from future survival. For a cell with a stabilizer, uniqueness also enforces stabilizer invariance. Apply the construction on cell orbits and transport it; repeated representatives in a full orbit merely repeat the same averages.

Boundary of the improvement. This proves admissibility for the local cap, orbit, endpoint-capacity and same-mask matching requirements. It does not prove a smaller final broad energy, a smaller total deletion, or an improved global theorem. Later coarsening and future histories depend on the chosen weights. Transferring the full arithmetic construction requires its uniform counting inputs to apply to the resulting histories; those inputs are imported here. Even with that transfer, a greedy minimum of J is not a theorem about the minimum of all future $J + K'$ costs.

Proposition 4.3 (A strict source-compatible local improvement)**.** *Use one mask, $\pi = 1$, two plus and two minus entries of raw weight one, and $N_0 \geq 2$. Put plus i and minus i in class A_i, for $i = 1, 2$, and choose distinct colors. Product coupling has $x_{ij} = 1/2$ and $J = r/p$. The diagonal matching $x_{11} = x_{22} = 1$ has $J = 0$.*

Proof. For product coupling, the two class increments are respectively $(S_1 - S_2)/2$ and its negative. Since the colors differ, $\mathbb{E}(S_1 - S_2)^2 = 2/p$, so the sum of their variances times r is r/p. For diagonal matching, each class retains exactly its one matched unit, at the plus endpoint on success and at the minus endpoint on failure. Both class masses are identically one. Both plans have the same total matching and obey all endpoint and cell caps. This is a manufactured finite cell, not a claim that a specific source arithmetic block realizes it. □

4.1 Exact certificates and computational cost

Set $a_{c,A}(X) = (z_c - \bar{z})_A$. Differentiating the objective gives

$$g_{ij}(X) = \frac{\partial J}{\partial x_{ij}} = -\frac{2r\pi_h}{p} a_{c_i, A_j^-}(X). \tag{10}$$

The derivative of $\bar{z}$ disappears because $\sum_c a_{c,A} = 0$.

Proposition 4.4 (An independently checkable optimality witness)**.** *A feasible X_* minimizes J if and only if*

$$\sum_{ij} g_{ij}(X_*)(x_{ij} - x^*_{ij}) \geq 0 \quad \textit{for every feasible } X. \tag{11}$$

One sufficient finite certificate is, for each mask, nonnegative row and column prices α_i, β_j and a real τ such that

$$\begin{aligned} g_{ij} + \alpha_i + \beta_j - \tau \geq 0, \qquad & x^*_{ij}(g_{ij} + \alpha_i + \beta_j - \tau) = 0, \\ \alpha_i(u_i - \sum_j x^*_{ij}) = 0, \qquad & \beta_j(v_j - \sum_i x^*_{ij}) = 0. \end{aligned} \tag{12}$$

For rational incoming data and a rational proposed witness all these equalities and inequalities can be checked with rational arithmetic.

Proof. Writing $a(X) = a(X_*) + a(X) - a(X_*)$ gives the exact expansion

$$J(X) - J(X_*) = g(X_*) \cdot (X - X_*) + \frac{r}{p} \|a(X) - a(X_*)\|_2^2. \tag{13}$$

Thus (11) suffices. Necessity follows by taking the right derivative along every feasible segment from a minimizer. For (12), nonnegativity and the capacity bounds yield $g \cdot X \geq \tau \min(P_h, M_h) - \sum_i \alpha_i u_i - \sum_j \beta_j v_j$. Complementarity makes this an equality for X_*, proving the certificate. □

The zero-cost criterion is a linear feasibility problem; otherwise the primary problem is a convex quadratic program with at most the number of allowed plus–minus pairs as variables. It is not claimed cheaper than product coupling. One can aggregate equal-color fluxes without enumerating all p colors: if only k colors occur, the $p-k$ empty colors all have $z_c = 0$ and contribute $(p-k)\|\bar{z}\|^2$ to the sum. Floating-point optimizer output is not an exact certificate. The rational witness test or a rigorously bounded gap must accompany it. No universal runtime bound for the enormous source prime ranges is asserted.

5 Sharper finite loss accounting

Proposition 5.1 (Load-sensitive clipping)**.** *In one mask and new cell let total incoming raw load be x, with old child contributions $x_D \le N_0$ and additional newborn load. Set $C = x^2 - \sum_D x_D^2$. If $x > N_0$, proportional clipping loses exactly $x - N_0$ and*

$$x - N_0 \le C/x \le C/N_0. \tag{14}$$

The first bound is attained when the cell consists of an integer number greater than one of saturated old-child loads and no newborns. The first-mass multiplier for the entire inequality is $r\pi_I(a)$.

Proof. Since $\sum_D x_D \le x$, we have $\sum_D x_D^2 \le N_0 \sum_D x_D \le N_0 x$. Thus $C \ge x(x - N_0) \ge 0$, and division gives the first bound; the second uses $x > N_0$. For k saturated children, $x = kN_0$ and $C = k(k-1)N_0^2$, giving equality. The common-mask first mass is linear in $\pi_I(a)$; no second probability factor is introduced. □

This is a per-cell refinement, not a uniform improvement of the source's $1/N_0$ bound when loads are unknown. It becomes useful when the actual loads are retained in the certificate. No new inverse estimate is inferred.

Proposition 5.2 (Loss-sensitive accounting-energy budget)**.** *Consider a finite chronology of births, mass-preserving prime updates, class coarsenings and deletions. All class masses at deletion times and at the end are at most N_0, each class has fixed width r_A, and restrictions never split classes. Let W_b be total inserted first mass, B total inserted accounting energy, and $L = \sum_{t,A} r_A d_{t,A}$ total discarded first mass. At a deletion, let $m_{t,A}$ and $d_{t,A}$ be its pre-mass and removed mass. Then*

$$\begin{aligned} \sum_t (J_t + K'_t) &= \mathcal{A}_{\mathrm{end}} - B + \sum_{t,A} r_A (2 m_{t,A} d_{t,A} - d_{t,A}^2) \\ &\le N_0(W_b + L) - B - \sum_{t,A} r_A d_{t,A}^2. \end{aligned} \tag{15}$$

In particular this is at most $2N_0 W_b$, since $L \le W_b$.

Here every scalar is a deterministic per-unit spatial sum over complete translation orbits, as in the source. In particular J_t is the actual full-orbit energy increase represented by the orbit-variance identity; it is not the energy change of one chosen random translated cell. First-mass preservation is required for these complete sums. Alternatively, for a genuinely random chronology, read every scalar term in (15) under $\mathbb{E}$, and require first-mass preservation in expectation. Mean preservation never implies pathwise preservation.

Proof. Deleting d from mass m lowers its square by exactly $2md - d^2$. Births inject B, updates inject J_t and coarsenings inject K'_t. Telescoping gives the equality, starting from the empty table. Final first mass is $W_b - L$, whence $\mathcal{A}_{\text{end}} \le N_0(W_b - L)$. Also $m_{t,A} \le N_0$. Substitution proves the inequality. No loss from splitting a class has been silently discarded, because splitting was forbidden. □

The source already proves an $O(N_0 W)$ variance budget. The exact identity and loss-dependent bound above refine its bookkeeping; they are not credited as a separate global theorem. Static losses omitted before a birth should not be counted again in L. In a source application $W_b \le W$, while the virtual-loss ledger additionally restores the appropriate omitted birth losses.

6 Exact interval moments and common-law deletion

Let I_i be a circle interval of full length $r_i < 1/2$ centred at x_i. For a finite table history ω let $a_i(\omega) \ge 0$ be its final weight, zero for absent labels. Define $F_\omega(x) = \sum_i a_i(\omega)\mathbf{1}_{I_i}(x)$. Assume $a_i \in L^2(\Omega)$ for every label; bounded weights in a fixed finite source block satisfy this automatically. The spatial row length is r_i, not $2r_i$: the source's retained indicator has half-width $r_i/2$ inside the original approximation interval. Write $K_{ij} = m(I_i \cap I_j)$ and $\ell_i = r_i$.

Proposition 6.1 (Exact finite interval Gram formulas)**.** *On the common probability space* $(\Omega \times \mathbb{T}, \mathbb{P} \otimes m)$*,*

$$\mathbb{E}\int F = \sum_i \ell_i \mathbb{E}a_i, \qquad \mathbb{E}\int F^2 = \sum_{ij} K_{ij}\mathbb{E}(a_i a_j), \tag{16}$$

$$\int (\mathbb{E}F)^2 = (\mathbb{E}a)^T K (\mathbb{E}a), \qquad \mathbb{E}\int F^2 - \int (\mathbb{E}F)^2 = \mathbb{E}[(a - \mathbb{E}a)^T K (a - \mathbb{E}a)] \ge 0.$$

Both K and $K - \ell\ell^T$ are positive semidefinite. For fixed coefficient vectors $0 \le a \le b$, with $d = b - a$, the exact deletion energy is $a^T K a = b^T K b - 2b^T K d + d^T K d$.

Proof. The functions $\mathbf{1}_{I_i}$ are square integrable on a probability space. Their uncentred and centred Gram matrices are K and $K - \ell\ell^T$, respectively. Expand the finite sums and integrate. Subtracting the coefficient mean gives the second line, and substituting $a = b - d$ gives the last assertion. No cross term has been suppressed. □

For exact rational endpoints, split every interval at zero, sort all endpoints, and use the common half-open atoms A_s of lengths t_s. If $H_{si} = \mathbf{1}_{A_s \subset I_i}$, then

$$K = H^T \operatorname{diag}(t_s) H. \tag{17}$$

An independent entry calculation sums $\max(0, \min(b, d) - \max(a, c))$ over the split pieces $[a, b)$ and $[c, d)$ of two intervals. Endpoints have measure zero. These two calculations provide a finite check of all overlaps, including wrap-around and coincidences. For non-rational γ an ordering or interval-arithmetic certificate is needed; rational fixtures do not certify arbitrary real input comparisons.

The source's broad distance kernel is a nonnegative entrywise majorant for K, not a matrix asserted here to be positive semidefinite. Indeed $K_{ij} \le \min(r_i, r_j)$, and it vanishes unless $\operatorname{dist}(x_i, x_j) \le (r_i + r_j)/2 \le \max(r_i, r_j)$. Thus its broad-energy bound controls the retained L^2 norm.

Two distinct averaging operations. Static anchor-averaged coefficients are already coefficients of a realized table. Their products use independent copies of the auxiliary anchors conditional on that table. In contrast, both factors $a_i(\omega)a_j(\omega)$ in (16) use the same realized table history ω. Replacing that expectation by $\mathbb{E}a_i\mathbb{E}a_j$ computes $\|\mathbb{E}F\|_2^2$, not $\mathbb{E}\|F\|_2^2$. Neither operation randomizes γ.

6.1 Realized fields, not recomputed plans

For one true-grid label pulled through its ancestors, let $f_j \in [0,1]$ be the actual nondiscard multiplier and $d_j \in [0,1]$ its retained discard fraction. With the hard birth coefficient suppressed, the exact identity is

$$\prod_{j=1}^{n} f_j - \prod_{j=1}^{n}(f_j d_j) = \sum_{j=1}^{n}\left(\prod_{k<j} f_k d_k\right) f_j(1-d_j)\left(\prod_{k>j} f_k\right). \tag{18}$$

It follows by removing the discard factors successively, or by induction. Each term has the actual prefix and the virtual suffix. The source removes future fields by reverse orbit averaging, and recovers $R(I/a)/a = \pi_I(a)$ at the discard time. These symmetries are an additional source fact, not a consequence of the product identity.

For the optimized history, the same identity uses the optimized history's realized f_j. It must not recompute a second set of optimized plans on deletion-free arrays. Such a recomputation changes f_j, destroys this telescoping comparison and need not preserve domination. The sum of *all* restored expected losses, including static, soft, clipping and partition losses, is what enters the transfer theorem.

7 A weaker analytic transfer interface

Let $G_n = \mathbb{E}_\omega F_{n,\omega} \geq 0$ on $\mathbb{T}$. We use the absolute continuity modulus

$$\omega_G(t) = \sup_n \sup_{m(A)\leq t} \int_A G_n \, dm, \qquad 0 \leq t \leq 1. \tag{19}$$

For a uniformly L^1-bounded nonnegative family on a probability space, $\omega_G(t) \to 0$ is equivalent to uniform integrability. For completeness, a uniform tail bound gives $\int_A G_n \leq Km(A) + \int_{G_n>K} G_n$. Conversely, if $\sup_n \int G_n \leq M$, then $m(G_n > K) \leq M/K$ and the tail integral is at most $\omega_G(M/K)$. These inequalities prove the equivalence.

Theorem 7.1 (Uniform-integrability finite-block transfer)**.** *Fix a measurable set $E \subset \mathbb{T}$ with $m(E) = \mu > 0$. Suppose that, after choosing constants and loss budgets, arbitrarily late finite blocks admit measurable arrays $0 \leq F_{n,\omega} \leq Z_{n,\omega}$ with the following properties. There are constants $0 < m_0 \leq M_0 < \infty$ such that $V_n = \mathbb{E}\int Z_n \in [m_0, M_0]$ and*

$$\mathbb{E}\int (Z_n - F_n) \leq \eta V_n + b, \qquad 0 \leq \eta \leq \mu/8, \quad 0 \leq b \leq \mu m_0/8. \tag{20}$$

For every fixed finite union of intervals U there are numbers $e_n(U) \to 0$ with

$$\int_U \mathbb{E}Z_n \geq (m(U) - e_n(U))V_n. \tag{21}$$

Finally $\omega_G(t) \to 0$ for $G_n = \mathbb{E}F_n$. Then $F_{n,\omega}$ cannot vanish almost everywhere on E for every sufficiently late block and almost every ω.

Proof. Choose $\delta > 0$ with $\delta \le \mu/8$ and $\omega_G(\delta) < \mu m_0/4$. By regular approximation of Lebesgue measurable sets, fix a finite union U with $m(U \triangle E) \le \delta$. This choice is made after m_0, the loss budgets and the modulus are fixed. Only now take a sufficiently late block that $e_n(U) \le \mu/8$. Then $m(U) \ge 7\mu/8$, and the two premises imply

$$\int_U G_n \ge (3\mu/4 - \eta)V_n - b \ge \mu m_0/2.$$

If $G_n = 0$ almost everywhere on E, the same integral is at most $\int_{U\setminus E} G_n \le \omega_G(\delta) < \mu m_0/4$, a contradiction. Nonnegativity and Tonelli show that the stated vanishing of F_n implies that of G_n. □

Corollary 7.2 (Conditional limsup deduction)**.** *For the fixed γ, ψ in* (1)*, suppose that for every hypothetical positive-measure tail-avoiding set E, arrays satisfying Theorem 7.1 can be constructed with $F_{n,\omega}$ supported in unions of approximation intervals from arbitrarily late denominators. Then* $m(\limsup E_q) = 1$.

Proof. The complement of the limsup is $\bigcup_N \bigcap_{q\ge N} E_q^c$. Positive measure would give a positive-measure member of this countable union. All sufficiently late retained arrays vanish on that set, contrary to Theorem 7.1. □

The source supplies a uniform retained L^2 bound conditional on its arithmetic inputs, so it supplies the required modulus by Cauchy–Schwarz. The theorem weakens that analytic interface, not those arithmetic inputs. It also permits an L^p bound for any fixed $p > 1$:

$$\sup_n \|G_n\|_p \le C_p \quad \Longrightarrow \quad \omega_G(t) \le C_p t^{1-1/p}. \tag{22}$$

Hölder proves this directly. Alternatively, a bound $\sup_n \int \Phi(G_n) \le B$ for a nonnegative function with $\Phi(s)/s \to \infty$ gives $\omega_G(t) \le Kt + B \sup_{s>K} s/\Phi(s)$. No convexity is needed for this sufficient inequality.

Strictness of the analytic weakening. On $(0, 1]$ let $G_n(x) = \min(n, x^{-3/4})$. These are dominated by the integrable function $x^{-3/4}$ and $\omega_G(t) \le 4t^{1/4}$, by decreasing rearrangement (or by maximizing the integral on an initial interval). Their squared integrals grow without bound: on $[n^{-4/3}, 1]$ the integral of $x^{-3/2}$ is $2(n^{2/3} - 1)$. Thus uniform integrability need not give a uniform L^2 bound. This fixture illustrates only the analytic premise, not an arithmetic construction of such arrays.

In contrast, bounded first mass alone is insufficient: $G_n = n\mathbf{1}_{[0,1/n]}$ has integral one and $\omega_G(t) = 1$ for every $t > 0$. Equidistribution is independently necessary: constant repetition of an interval-supported test of length $1/2$ has bounded L^2 norm and positive mass but avoids the other half. No quantitative denominator rate follows from the fixed-U convergence in (21). A moving U_n, or optimizing its complexity with n, needs a new uniform discrepancy estimate.

8 Verification, limitations, and research directions

The independently executable file `M048-checks.py` uses exact integer and rational arithmetic. It compares direct residue enumeration with flux and covariance formulas, checks endpoint means and caps, tests the strict two-class family and quadratic certificate expansion, checks one-law clipping

and later-omission normalization, and compares two independent interval-overlap calculations with common-law moments. It includes a finite product-telescoping test and counterexamples to independence and to replacing the expected square by the square of the expectation. The actual counts and input ranges are recorded in `M048-checks.json`; none are parent-run counts. The completed author run passed 44,135 assertions, including 480 two-mask orbit fixtures, 783 two-class plans, 742 clipped cells, 240 deficit laws, 160 circle-interval tables and 320 discard products.

All finite fixtures are manufactured. They neither construct the source's astronomical arithmetic tables nor check prime-factor extraction, dense-bin purges, rational-model exclusions, uniform sparse counts, every chronological induction, or the mass-transference theorem. Exact finite checks supplement the displayed proofs; they cannot prove an infinite statement by exhaustion. Every final PDF page is visually inspected, separately from mathematical checks and source-file hashing.

8.1 Corrections and details that must not be lost

No substantive error is established in the inspected public source passages. The report does not issue a verdict on the unread proof complements. In particular, the source already correctly keeps:

- one outer mask factor in clipping and capacity charges, versus two in broad pairs, even on the diagonal;
- simultaneous residue translations rather than independent success tests, and realized adaptive fields in virtual products;
- all restored expected deletion losses, with soft loss measured relative to the actual virtual mass, not divided again by its lower bound;
- inherited classes after a connecting label vanishes, and the fixed parameter–fixed union–late denominator order.

These are retained valid details, not corrections newly discovered here. The load-sensitive clipping coefficient, exact loss budget, optimizer and UI interface are qualified extensions of those details. The budget's averaging convention is explicit so a mean-preserving cell experiment is not confused with a pathwise conservation law.

8.2 Benefits, costs, and feasible next steps

The optimized matching preserves the source's elementary local invariants and never increases the next accounting variance for the same incoming data. Its zero-cost test and dual witness make the improvement independently auditable. The exact interval Gram exposes overlap and dependence errors; the UI theorem identifies the weakest analytic assumption used by this particular fixed-set contradiction. These are useful without crediting a new global Diophantine theorem.

The costs are real. Joint optimization across masks can be much larger than product coupling, which has a closed form. Accounting variance is an auxiliary potential, not the eventual broad energy. Rational fixtures do not settle real-number ordering, and finite positivity does not furnish small arithmetic correlations. No priority claim is made for covariance, convex optimization, Hölder, or uniform-integrability arguments.

Three bounded research directions follow. First, identify incoming source configurations where the zero-flux feasibility problem has a structural solution and certify it without materializing

all residue colors. Second, test whether minimizing a combination of present flux and explicitly bounded future coarsening cost yields a provable multi-step gain; an unchecked greedy extrapolation is insufficient. Third, seek arithmetic tail estimates for $\mathbb{E}F_n$ strong enough to give a UI modulus when a full second moment is expensive. Any such estimate must be uniform after the loss parameters are fixed, and must preserve the entire expected-deletion comparison. The present report does not supply it.

Assessment. Applicability remains L3. C3 refers to the proved, strictly improving finite matching construction and the conditional analytic weakening, not to relabeling a Gram matrix or to validation of the source's global claim. The assessment is stored in the required list schema with literal theorem labels and the final TeX hash. The owned freeze binds the final report, tests, read/proof/visual evidence and build outputs, not mutable parent integration records.

References

[1] OpenAI mathematics collection, *The Weak Inhomogeneous Duffin–Schaeffer Conjecture*, September 25, 2026. Cached public source at pinned commit `adc7f1241b42e322a6451854ab7e4b4c146bf78a`; public source directory. Scoped reading, not full proof certification.

[2] D. Koukoulopoulos and J. Maynard, *On the Duffin–Schaeffer conjecture*, Annals of Mathematics 192 (2020), 251–307. `https://annals.math.princeton.edu/2020/192-1/p05`. Publisher statement inspected; complete proof not reread.

[3] V. Beresnevich, M. Hauke and S. Velani, *Borel–Cantelli, zero-one laws and inhomogeneous Duffin–Schaeffer*, arXiv:2406.19198v1 (2024), Theorems 11, 12, 14. `https://arxiv.org/html/2406.19198v1`. Primary statements inspected October 8, 2026; not a complete proof audit.

[4] M. Hauke-Treuer, J. Maynard and A. Pollington, *Counterexamples to the inhomogeneous Duffin–Schaeffer Conjecture*, arXiv:2609.30921 (September 25, 2026). `https://arxiv.org/abs/2609.30921`. Recent preprint claim; abstract/primary statement scope only.

[5] Y. He and L. Liao, *Counterexamples to the inhomogeneous Duffin–Schaeffer conjecture for a residual set of shifts*, arXiv:2609.30870v1 (September 25, 2026), Theorems 1–2. `https://arxiv.org/html/2609.30870v1`. Primary statements inspected; full proofs not audited.

[6] M. Subbarao, *Joint-Moment Bayesian Neural Inference and Inverse Uncertainty Propagation*, TR-core-pat-16, October 1, 2026. Confidential author-supplied original, selected complete covariance and remainder proofs; exact local path and intervals in the private ledger.

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